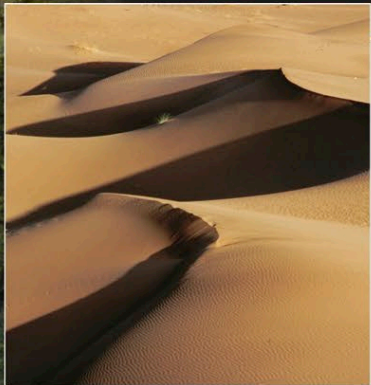




RAINFOREST

Forests cover 30 percent of Earth's land surface and range from the cold, dark boreal forest of the far north to the dense rainforests of the humid tropics.



SANDY DESERT

Deserts cover about 20 percent of Earth's land surface, but only a small proportion are occupied by sand dunes, like these in the Sahara Desert.

SURFACE FEATURES

From space, the flatter areas of Earth's land surface (apart from the areas dominated by ice) appear either dark green or various shades of yellow-brown. The green areas are forests and grasslands, which comprise a major component of Earth's biosphere (the planet's life-sustaining regions). The yellow-brown areas are mainly deserts, which have been created over long periods by various weathering and erosional processes. Like the other rocky planets, Earth has suffered many thousands of meteorite impacts over its history (see p.103). But because Earth's surface is so dynamic, the evidence for most of these impacts has disappeared, removed by erosion or covered up by depositional processes.

WATER

Water is a dominant feature of Earth's surface. Overall, about 97 percent of the water is in oceans (which cover 75 percent of the surface); 2 percent is in ice sheets and glaciers; less than 1 percent is in ground water (underground and in rocks); and the rest is in rivers, lakes, and the atmosphere. The presence of liquid water has been key to the development of life on Earth, and the heat capacity of the oceans has been important in keeping the planet's temperature relatively stable. Liquid water is also responsible for most of the erosion and weathering of Earth's continents, a process unique in the solar system, although it is believed to have occurred on Mars in the past.

KINGDOMS OF LIFE

Biologists use various systems for classifying living organisms, but the most widely used is the five-kingdom system. This classifies organisms mainly on the basis of their cell structure and method of obtaining nutrients and energy. However, not all scientists accept this system as satisfactory, and some have proposed switching to an eight-kingdom system or one with 30 kingdoms grouped into three superkingdoms.

ANIMALS

Animals are multicellular organisms that contain muscles or other contractile structures allowing some method of movement. They acquire nutrients, so they gain energy by ingesting food. Many animals, including mammals, are vertebrates (they possess a backbone), but a far larger number are invertebrates (without a backbone).

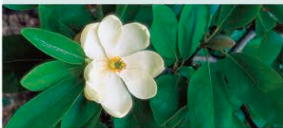
VERTEBRATE



PLANTS

Plants are multicellular organisms that obtain energy from sunlight through the process of photosynthesis. Their cells contain special pigments for absorbing light energy and are enclosed by cell walls made of cellulose.

FLOWERING PLANT



FUNGI

Fungi acquire nutrients by absorption from other living organisms or dead and decaying organic material. They have no means of locomotion. They range from yeasts (microscopic unicellular organisms) to multicellular forms with large fruiting bodies, such as mushrooms.

TOADSTOOL



PROTISTS

Protists are microscopic, mainly single-celled organisms whose cells contain nuclei. Some gain energy from sunlight, others ingest food like animals.

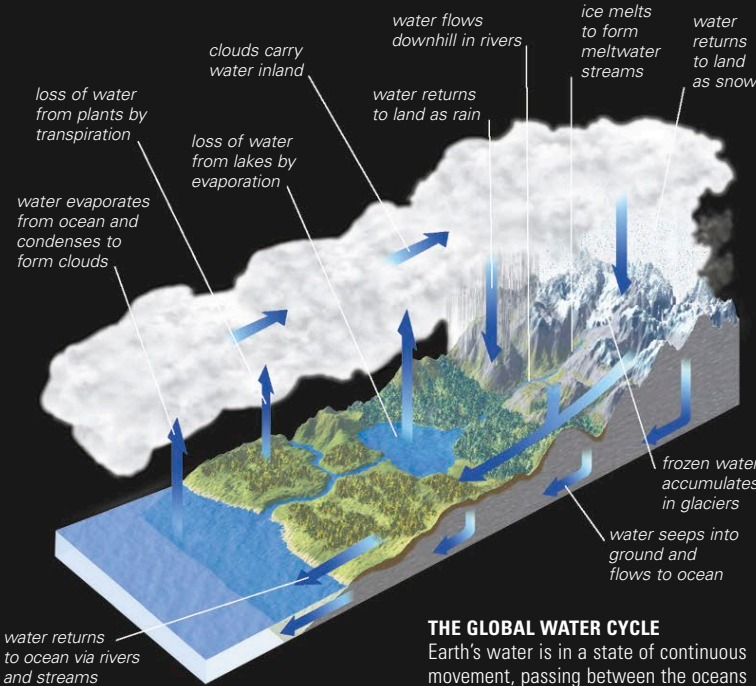
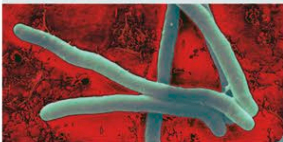
PARAMECIUM



MONERANS

Monerans are the simplest, smallest, most primitive, and most abundant organisms on Earth. The two main groups are bacteria and blue-green algae (cyanobacteria). Monerans are single-celled, but their cells contain no distinct nucleus. Most reproduce by splitting in two.

MYCOBACTERIUM



THE GLOBAL WATER CYCLE

Earth's water is in a state of continuous movement, passing between the oceans and lakes, the atmosphere, and the land in a cycle of connected processes.

LIFE ON EARTH

Evidence in ancient rocks points to the presence of simple, bacteri-like organisms on Earth some 3.8 billion years ago. However, the prevailing scientific view is that life started on Earth long before that as a result of complex chemical reactions in the oceans or atmosphere. These reactions eventually led to the appearance of a self-replicating and self-repairing molecule, a precursor of DNA (deoxyribonucleic acid). Once life in this rudimentary form had started, processes such as mutation and natural selection inevitably led, over the vast expanses of geological time, to a collection of life forms of increasing diversity and complexity. Life spread from the oceans to the land and to every corner of the planet. Currently, Earth is teeming with life in astonishing abundance and diversity.